By Daniel Mattioli

Data Capture Technologies – RFID Door Locking System

Problem statement & hypothesis

RFID door locking systems are pivotal for safety and security in secure settings such as office buildings and smart homes. However, current systems have limitations such as advanced security. They hinder security personnel's ability to review unauthorized access attempts as well as restrict integration with other smart home features for automation.

By enhancing RFID systems with a real-time clock and data logging capabilities, we can Enhance security by providing detailed access records as well as Improve smart home experiences through predictive adjustments based on entry/exit patterns.

However, key questions arise with these improvements such as how can this data improve security and predict user behaviours effectively and how might inaccuracies impact user experience?

Approach

This project aims to improve security by logging all access attempts, simplifying the identification of users and access times. When an RFID card is near the scanner, a servo motor either unlocks the door or denies entry based on the card. Every scan logs the date, time, card ID, and access status on an internal storage device.

Ethical concerns

Using the RFID scanner system to collect data will present lots of ethical concerns around user privacy, data security and data retention. Each swipe of an RFID car will be recorded, essentially tracking an individual's movement. This data collection process brings up several ethical concerns.

Data Transparency – Users of the system should be aware and in an agreement about why their data is being logged.

Storage vulnerability – Storing user's data on an SD card allows criminals to easily have access to this locally stored data, thus also risking the user's privacy.

Potential misuse – if the SD card falls into the wrong persons hand, such as a cyber-criminal, the data can be weaponized in harmful ways.

Relevance of old data – The longer the data is retained in the SD card, the longer it poses a potential privacy risk to the users. However, this data may not be relevant to them anymore, and what was accurate a year ago, may not be the case now.

Technology & Connections used

1. RFID module: This reads the RFID tag number.
2. Microcontroller(Arduino Uno): Processes RFID data and manages components. Runs compiled code.
3. Data Logging Shield: Attaches to Arduino for SD card data capture.
4. Real-Time Clock: Timestamps each entry.
5. Servo Motor: Moves door lock pin.
6. Breadboard: Enables temporary circuits for Arduino Uno.
7. Jumper Wires: Connects components with signals.
8. LED Lights: 1 blue, 1 red. Flash based on card scan.
9. Buzzer: Provides audible feedback on access type.

RFID Module:

1. SDA → Arduino Digital Pin 2
2. SCK → Arduino Digital Pin 13
3. MOSI → Arduino Digital Pin 11
4. MISO → Arduino Digital Pin 12
5. GRN → Arduino Ground Pin
6. RST → Arduino Digital Pin 9
7. 3.3V → Arduino 3.3v Pin

Servo Motor:

1. Signal → Arduino Digital Pin 5
2. Power → Breadboard 5v Rail
3. Ground → Breadboard Ground Rail

Breadboard:

1. 5v & Ground extended on breadboard.
2. Buzzer: Positive → Arduino Pin 8; Negative → Breadboard Ground.
3. LED: Positive (via Resistor) → Arduino Digital Pins 6 & 5; Negative → Breadboard Ground.

Real world picture of the system

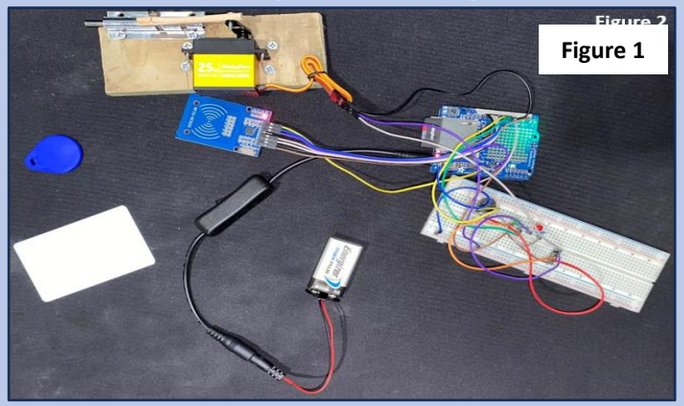
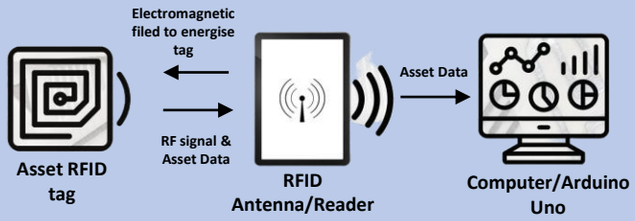


Figure 1



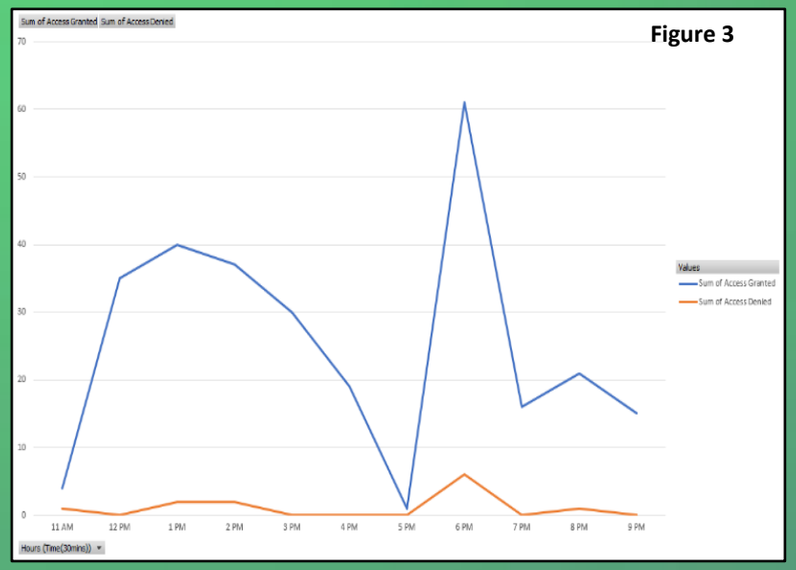


Figure 3

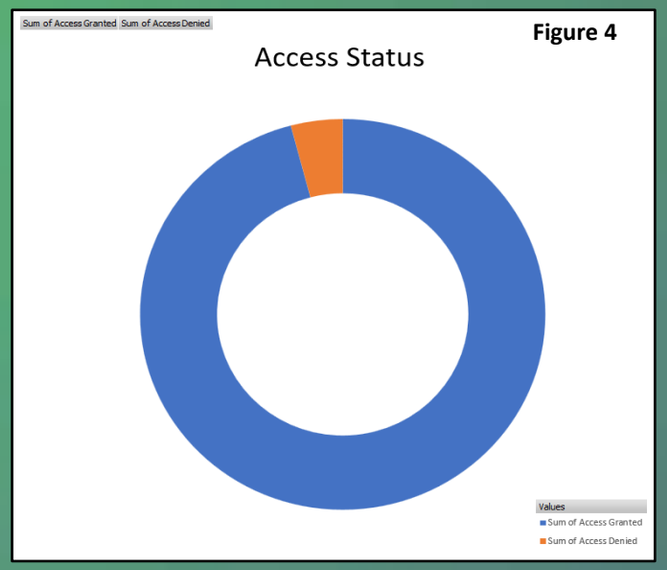
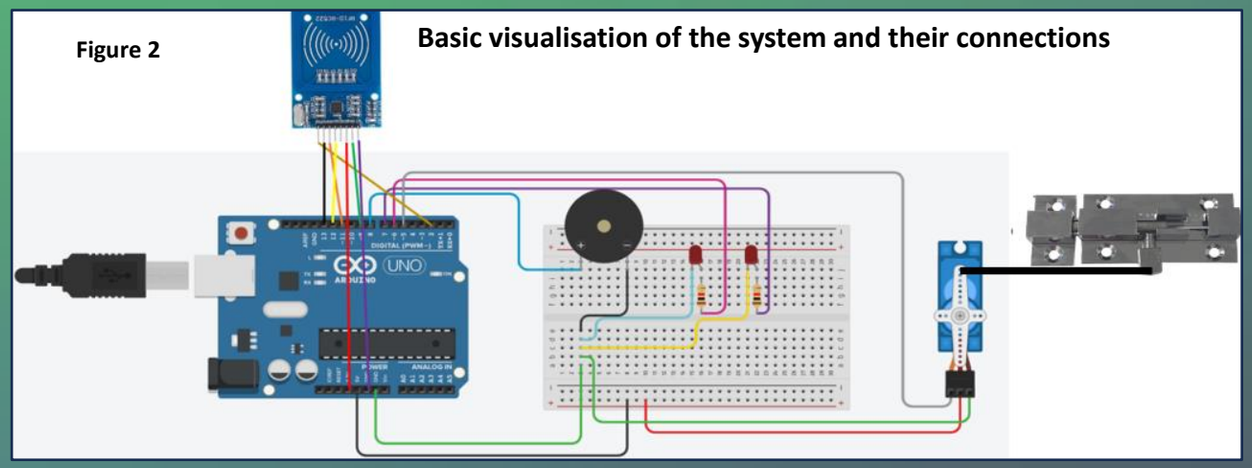


Figure 4

Figure 2

Basic visualisation of the system and their connections



Data collection

Data collection starts with the RFID card being scanned which will detect the serial number associated with it. My code has been setup to only grant access to a specific serial number. Each scan is then timestamped using the Real-Time Clock module. It is formatted and then written into a csv file stored on an SD card on the Arduino Uno.

Maintaining clean data is crucial for accuracy of the data. A delay was introduced into the code to stop duplicate entries. If duplicates were still added, excels "remove duplicates" tool can be used. Manual entries might introduce errors, so it's best to avoid them.

Challenges & Lessons learnt

Hardware integration such as combining components such as the RFID scanner and servo posed challenges. An improperly formatted SD card caused data-saving errors, and finding the right battery for the energy-hungry servo motor was tricky. Lessons learned include the importance of regularly checking SD card logs and valuing feedback from educators for swift problem-solving.

Results

The system was tested in a controlled setting, logging every RFID scan, including details such as the data, time, card ID and access status onto the Arduino Uno's SD card. Stored data was also kept uneditable, ensuring authentic data without any issues. Analysis showed a higher frequency of "Access Granted" compared to "Access Denied" attempts, mostly due to the correct key usage. Unauthorized access attempts were easily identifiable, ensuring potential security breaches could be promptly addressed.

While the current system lacks advanced analytics or smart home integration, the data sets the stage for potential integrations:

- Detected patterns, like common return times, can drive future smart home actions, given the system's scalability.
- Advanced software could better utilize the RFID and RTC modules for predictive analysis and home automation insights.
- Proposed hardware improvements such as an LCD screen for direct user feedback on access attempts
- Figure 3 displays the count of 'Access Granted' and 'Access Denied' attempts, highlighting peak activity times. Figure 4 visually contrasts successful and unsuccessful access attempts, providing a quick insight into their proportion.